Two Pictures of a Parity Flip: the Projector Involution $\Pi = P_1 - P_{23}$ and the Galois Conjugation of $\mathbb{F}_{169}/\mathbb{F}_{13}$

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September 26, 2026

Abstract

The ZETA architecture is a p-adic language model built entirely in $\mathbb{Z}_{13}[\eta]$ arithmetic: no floating point, no continuous phases, every operation carried out inside a fixed finite algebraic structure. A recurring question in that setting is what happens to the usual vocabulary of discrete symmetry, parity, time reversal, spectral involutions, once the underlying arithmetic is finite rather than continuous. This note answers that question for two order-two maps that already appear in the codebase and shows that they are, in a precise sense, the same map read at two different levels of the construction.

The first map is the operator parity involution $\Pi = P_1 - P_{23}$, built from a pair of complementary orthogonal projectors P_1, P_{23} acting on a channel space split into an address part and a content part. The second is the Galois conjugation $\sigma : \theta \mapsto -\theta$ of the quadratic extension $\mathbb{F}_{169} = \mathbb{F}_{13}(\theta)$ over its prime field $\mathbb{F}_{13}$. Both are involutions with a two-element spectrum, and both are illustrated here by a phase-line or lattice picture together with a three-dimensional reflection picture. We then read σ physically, as a discrete parity operator on the fixed line $\mathbb{F}_{13}$ and, one level down, as a discrete time-reversal operator on an order-eight phase generator built from θ: since -1 is already a square in $\mathbb{F}_{13}$, the honest imaginary unit is σ-fixed, and it is a square root of it, not i itself, that carries the time-reversal sign flip. Section 1 treats Π, Section 2 treats σ as a field automorphism, Section 3 develops the parity and time-reversal reading, and Section 4 records the common shape shared by both maps.

1 The parity involution $\Pi = P_1 - P_{23}$

Let P_1 and P_{23} be orthogonal projectors on a channel space V, satisfying

$$P_1 + P_{23} = I, \qquad P_1 P_{23} = P_{23} P_1 = 0, \qquad P_1^2 = P_1, \qquad P_{23}^2 = P_{23}. \tag{1}$$

Read P_1 as the projector onto the address channel and P_{23} as the projector onto the content channel of the same object. Define the parity operator

$$\Pi := P_1 - P_{23}. \tag{2}$$

Proposition 1. Π *is an involution:* $\Pi^2 = I$. *Moreover* $\Pi P_1 = P_1 \Pi = P_1$ *and* $\Pi P_{23} = P_{23}\Pi = -P_{23}$.

Proof. Using $P_1 P_{23} = P_{23} P_1 = 0$ and idempotence,

$$\Pi^2 = (P_1 - P_{23})^2 = P_1^2 - P_1 P_{23} - P_{23} P_1 + P_{23}^2 = P_1 + P_{23} = I. \tag{3}$$

For the second claim, $\Pi P_1 = P_1^2 - P_{23} P_1 = P_1$, and $\Pi P_{23} = P_1 P_{23} - P_{23}^2 = -P_{23}$; the same computations on the other side give $P_1 \Pi = P_1$ and $P_{23}\Pi = -P_{23}$. □

Corollary 1. Π *has exactly two eigenvalues on* V*:* $+1$ *on* $\operatorname{im} P_1$ *and* -1 *on* $\operatorname{im} P_{23}$*.*

This is the sense in which Π enacts a phase flip $s \leftrightarrow -s$: on the one-dimensional picture of its spectrum, the address channel sits at $+1$ and the content channel sits at -1, and Π swaps nothing between them since P_1 and P_{23} already occupy disjoint, complementary subspaces; it only records which side of the split a vector belongs to. Figure 1 shows this on a phase line (left) and, purely as a drawing aid, on a sphere whose two poles carry the two eigenvalues (right). The sphere is not a claim that the underlying system is a physical two-level system; it is only a convenient way to draw a pole-to-pole flip and to place the axis of Π vertically.

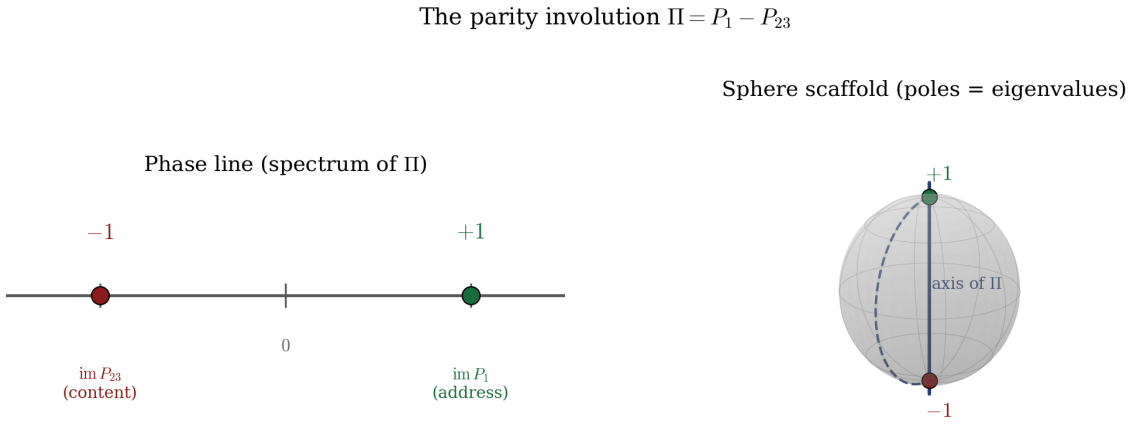


Figure 1: The parity involution $\Pi = P_1 - P_{23}$. Left: the two eigenvalues ± 1 on a phase line, labelled by the channel each one belongs to. Right: the same two eigenvalues placed at the poles of a unit sphere used only as a visual scaffold, with a single dashed pole-to-pole arc and a single solid vertical segment marking the axis of Π.

2 The Galois conjugation $\theta \leftrightarrow -\theta$ in $\mathbb{F}_{169} = \mathbb{F}_{13}(\theta)$

Let $d \in \mathbb{F}_{13}$ be a fixed quadratic nonresidue, and adjoin a root θ of $x^2 - d$ to form the field

$$\mathbb{F}_{169} = \mathbb{F}_{13}(\theta) = \{\, a + b\theta \;:\; a, b \in \mathbb{F}_{13} \,\}, \qquad \theta^2 = d. \tag{4}$$

This is the unique field of order 13^2, and it is a degree-two Galois extension of $\mathbb{F}_{13}$. Its Galois group is cyclic of order two, generated by the Frobenius map $x \mapsto x^{13}$, which acts on $\mathbb{F}_{169}$ as

$$\sigma(a + b\theta) = a - b\theta, \qquad a, b \in \mathbb{F}_{13}. \tag{5}$$

Proposition 2. σ *is an* $\mathbb{F}_{13}$*-algebra involution of* $\mathbb{F}_{169}$*: it fixes* $\mathbb{F}_{13}$ *pointwise, satisfies* $\sigma^2 = \mathrm{id}$*, and swaps the two roots of every irreducible monic quadratic over* $\mathbb{F}_{13}$ *with root in* $\mathbb{F}_{169} \setminus \mathbb{F}_{13}$*.*

Proof. For $a \in \mathbb{F}_{13}$, $\sigma(a) = \sigma(a + 0 \cdot \theta) = a - 0 \cdot \theta = a$, so σ fixes $\mathbb{F}_{13}$ pointwise. Applying σ twice, $\sigma^2(a + b\theta) = \sigma(a - b\theta) = a + b\theta$, so $\sigma^2 = \mathrm{id}$. If $\lambda = a_0 + b_0\theta$ with $b_0 \neq 0$ is a root of $x^2 - 2a_0x + (a_0^2 - b_0^2 d)$, direct substitution shows $\sigma(\lambda) = a_0 - b_0\theta$ is the other root. □

Concretely, take $\lambda_2 = 10 + 2\theta$. Its conjugate is $\lambda_3 = \sigma(\lambda_2) = 10 - 2\theta$, obtained by negating only the θ-component and leaving the $\mathbb{F}_{13}$-component untouched. Figure 2 draws this on the lattice $\{a + b\theta\}$ (left), where the fixed field $\mathbb{F}_{13}$ is the horizontal line $b = 0$, and again as a reflection in three dimensions (right), where the fixed field is drawn as a translucent plane and the vertical axis carries no arithmetic meaning; it is added only so the reflection is visible.

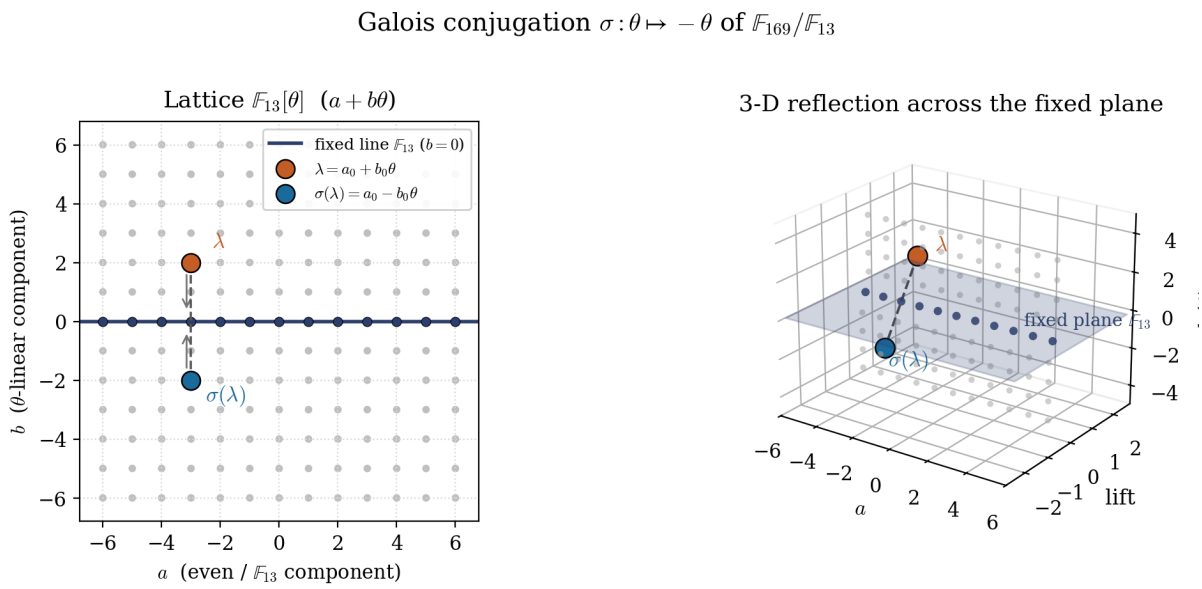


Figure 2: The Galois conjugation $\sigma : \theta \mapsto -\theta$ of $\mathbb{F}_{169}/\mathbb{F}_{13}$. Left: the lattice $a+b\theta$ with $a, b \in \mathbb{F}_{13}$, the fixed line $\mathbb{F}_{13}$ at $b = 0$, and a representative pair $\lambda, \sigma(\lambda)$. Right: the same reflection redrawn in three dimensions, with the fixed field as a plane and the vertical coordinate used purely to lift the picture off the page.

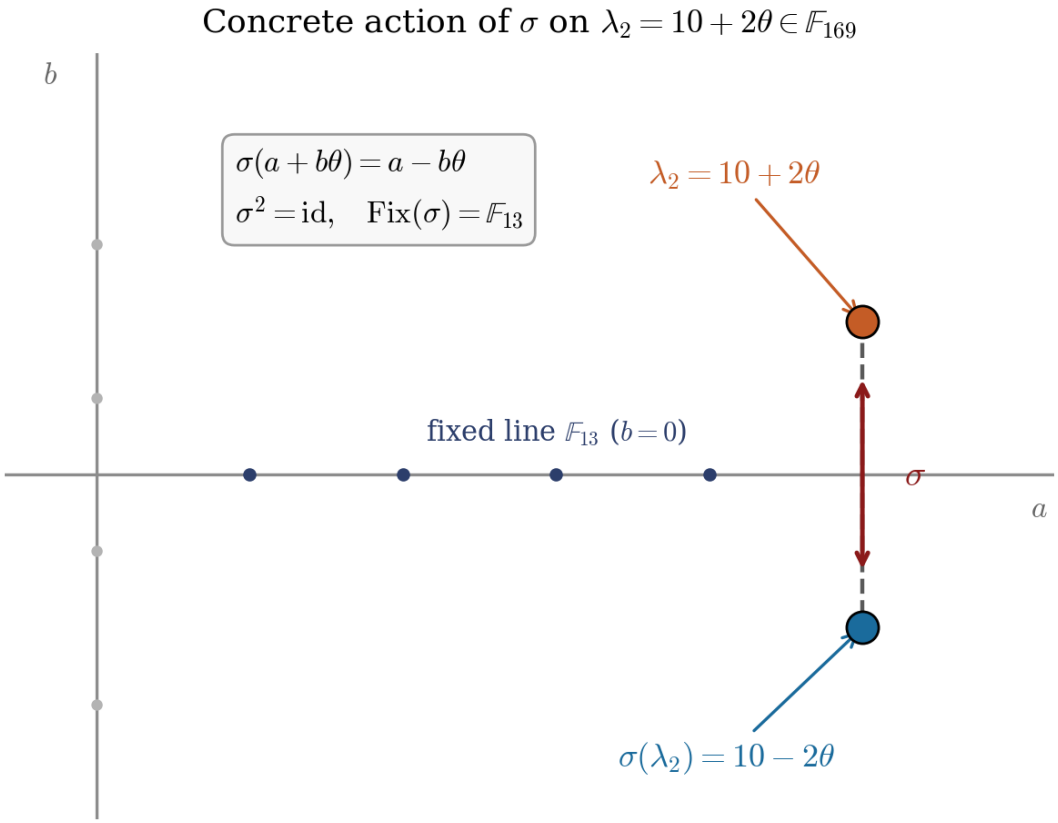


Figure 3: Concrete numerical action of σ on the element $\lambda_2 = 10 + 2\theta$. The map negates only the θ-component, leaving the $\mathbb{F}_{13}$-component fixed; the vertical dashed segment is the geometric realisation of the reflection across the fixed line $b = 0$.

3 Galois conjugation as a discrete parity and time-reversal involution

The instance of Proposition 2 used by ZETA fixes the nonresidue at $d = 2$, so that

$$\mathbb{F}_{169} \cong \mathbb{F}_{13}[\theta]/(\theta^2 - 2), \tag{6}$$

and $\sigma \in \mathrm{Gal}(\mathbb{F}_{169}/\mathbb{F}_{13})$ is the map $\sigma(a_0 + b_0\theta) = a_0 - b_0\theta$ of equation (5). Since σ generates the Galois group and $\sigma^2 = \mathrm{id}$ (Proposition 2), $\mathrm{Gal}(\mathbb{F}_{169}/\mathbb{F}_{13}) \cong C_2 \cong \mathbb{Z}_2$: it is, algebraically, nothing more than an order-two group acting on a state $\lambda = a_0 + b_0\theta$. What is worth recording is that this single involution carries two distinct physical readings at once, one on each grading of the field.

Discrete parity. Write $\lambda = a_0 + b_0\theta$ as an even part $a_0 \in \mathbb{F}_{13}$ and an odd, θ-linear part $b_0\theta$. Since θ plays the role of a pseudoscalar basis element, σ acts exactly as a parity inversion: it fixes the even sector pointwise and negates the odd sector,

$$\sigma(a_0) = a_0, \qquad \sigma(b_0\theta) = -b_0\theta, \tag{7}$$

which is the same even/odd split made precise in Section 4 below, and is drawn in Figure 2 as reflection across the fixed line $b_0 = 0$. Every state λ and its parity-flipped partner $\sigma(\lambda)$ sit on opposite sides of that line, so σ maps the two chiral sectors of $\mathbb{F}_{169} \setminus \mathbb{F}_{13}$ into one another.

Discrete time reversal. Time reversal on an honest complex Hilbert space is anti-unitary and, on a phase $e^{i\phi}$, acts as complex conjugation $i \mapsto -i$. A finite-field analogue of i is available here because $13 \equiv 1 \pmod 4$, so -1 is itself a quadratic residue in $\mathbb{F}_{13}$: concretely $5^2 = 25 \equiv -1 \pmod{13}$, so $i := 5 \in \mathbb{F}_{13}$ already satisfies $i^2 = -1$. Because $i \in \mathbb{F}_{13}$, it is fixed by σ along with the rest of the base field; the sign flip of time reversal is not visible on i itself, but one level down, on a square root of i that only exists after adjoining θ. Set

$$\mu := 3\theta, \qquad \mu^2 = 9\theta^2 = 9 \cdot 2 \equiv 5 \equiv i \pmod{13}, \qquad \mu^4 = i^2 \equiv -1 \pmod{13}, \tag{8}$$

so μ is an order-eight phase generator, a discrete eighth root of unity up to sign, whose square recovers the imaginary unit i. Since μ is θ-linear it lies entirely in the odd sector, and the parity computation above applies to it directly:

$$\sigma(\mu) = \sigma(3\theta) = 3\,\sigma(\theta) = -3\theta = -\mu. \tag{9}$$

This is an exact, zero-drift, finite-field realization of the anti-unitary sign flip $i \mapsto -i$: it is not i that is odd under σ, but the finer phase μ with $\mu^2 = i$, and $\sigma(\mu) = -\mu$ reproduces the time-reversal pattern without any continuous limit or floating-point phase.

The fixed subfield as invariant backbone. Both readings are consistent with the single fact already proved in Proposition 2:

$$\mathrm{Fix}(\sigma) = \{\, \lambda \in \mathbb{F}_{169} : \sigma(\lambda) = \lambda \,\} = \mathbb{F}_{13}. \tag{10}$$

Every scalar invariant of the theory, being built from $\mathbb{F}_{13}$-valued quantities such as a_0 and $i = 5$, survives σ untouched, while every θ-linear quantity, whether it is read as a chiral (parity-odd) label or as a phase generator μ with $\mu^2 = i$ (time-reversal-odd), is negated. Discrete parity and discrete time reversal are therefore not two separate structures bolted onto $\mathbb{F}_{169}/\mathbb{F}_{13}$; they are the same involution σ, read on two different odd quantities living in the same one-dimensional complement $\mathbb{F}_{13}\theta$.

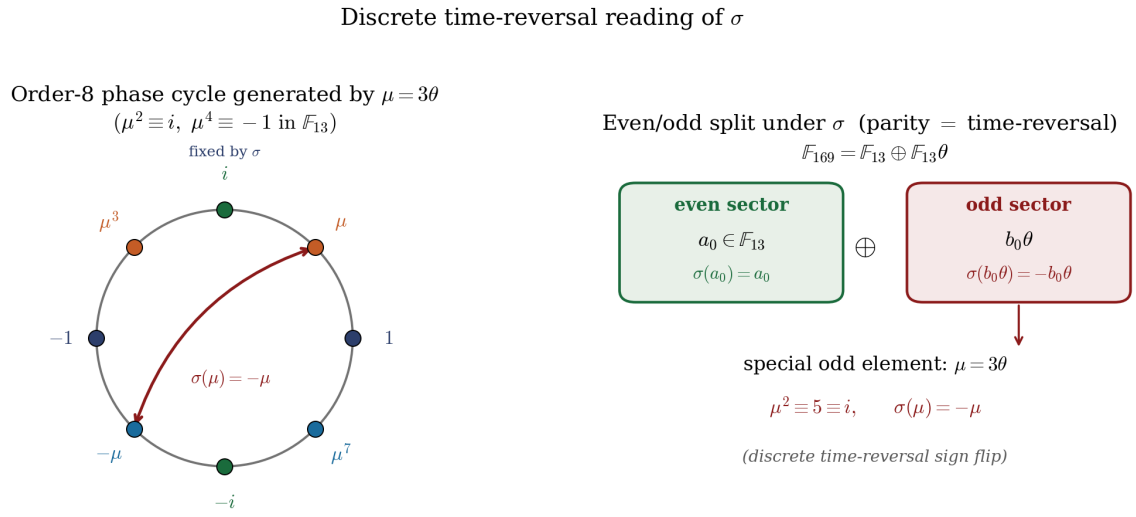


Figure 4: Discrete time-reversal reading of σ. Left: the order-eight phase cycle generated by $\mu = 3\theta$ (with $\mu^2 \equiv i$ and $\mu^4 \equiv -1$ in $\mathbb{F}_{13}$); the red arc marks the sign flip $\sigma(\mu) = -\mu$. Right: the even/odd decomposition $\mathbb{F}_{169} = \mathbb{F}_{13} \oplus \mathbb{F}_{13}\theta$, showing that σ acts as the identity on the even sector and as negation on the odd sector (the same involution that realises both discrete parity and discrete time reversal).

4 The common shape of both pictures

Both Π and σ are order-two maps built from a decomposition of an ambient space into a part that is preserved and a part that is negated:

- $\Pi = P_1 - P_{23}$ fixes $\operatorname{im} P_1$ pointwise (eigenvalue $+1$) and negates $\operatorname{im} P_{23}$ (eigenvalue -1), with $\operatorname{im} P_1 \oplus \operatorname{im} P_{23} = V$.
- σ fixes $\mathbb{F}_{13}$ pointwise and negates the θ-direction, with $\mathbb{F}_{13} \oplus \mathbb{F}_{13}\theta = \mathbb{F}_{169}$ as $\mathbb{F}_{13}$-vector spaces.

This is why the same two-panel layout, a flat picture of the two eigenvalues followed by a three-dimensional reflection-across-a-fixed-set picture, illustrates both maps: in each case the object of interest is an involution with a $+1$ eigenspace that is drawn as a fixed line or plane, and a -1 eigenspace that is drawn as the direction being reflected.